

SIVAMANI R

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PROFESSIONAL SUMMARY

AI Engineer with hands-on Python experience building Generative AI pipelines using OOP-based, modular code, structured prompt engineering and tool/function calling. Applies data science fundamentals preprocessing, exploratory analysis, and model evaluation with NumPy, Pandas, and Scikit-learn alongside LLM-driven backends .

SKILLS

GenAI: LLMs | Prompt Engineering | Tool Calling | LangChain | LangGraph | RAG | AI Agents | Hugging Face | Vector Databases | pgvector | Ollama | Ragas | Gemini | Groq
AI/ML & Deep Learning: TensorFlow | Keras | PyTorch | CNN | NLP | Computer Vision | OpenCV | Deep Learning
Data Science: Python | NumPy | Pandas | Scikit-learn | Matplotlib | SQL | Data Preprocessing | Statistics | EDA | Model Evaluation
Backend & Tools: FastAPI | Flask | REST APIs | Async Processing | Pydantic | Docker | Git | GitHub Actions | Postman

EXPERIENCE

AI Backend Intern | *Petals Automation Private Limited — Madurai, Tamil Nadu* **Feb 2026 – Apr 2026**

- Designed structured Generative AI prompt templates using React techniques to convert natural language into predefined program blocks, reducing manual configuration effort by 50–60%
- Implemented tool calling to connect LLM outputs with OOP-based, modular backend Python functions, improving execution accuracy and reducing invalid responses
- Built an async FastAPI backend with Pydantic validation, JWT-based authentication, and request logging to integrate LLM workflows
- Developed an AI agent workflow using LangGraph with controlled state transitions and a PostgreSQL-backed validation layer to enforce schema constraints, catching incorrect outputs across 20–40 test cases

Graduate Engineer Trainee | *Sun Mobility Private Limited — Bangalore, Karnataka* **Aug 2023 – Dec 2024**

- Cleaned, validated, and preprocessed EV battery datasets (5K–10K records per cycle), applying exploratory data analysis to build monitoring dashboards tracking efficiency, degradation, and cycle metrics for engineering-level reporting

AI / ML PROGRAM

Generative AI Query Agent | *Technologies: FastAPI | LangGraph | PostgreSQL | Google Gemini | Pydantic |* **Jul 2025 – Dec 2025**

- Built a multi-step Generative AI pipeline for text generation, reasoning, and response generation using structured prompt flows across input parsing and response stages, with conversation state stored in PostgreSQL
- Developed an async, OOP-structured FastAPI backend with streaming responses, JWT-based authentication, and request logging, supporting 10+ concurrent requests
- Applied Pydantic validation and structured error handling to manage invalid inputs and handle 20+ edge cases during testing

Legal Document Retrieval Using RAG | *Technologies: LangChain | OpenSearch | Llama 3.2 3B | Ollama | Groq | RAGAS |Hugging Face*

- Built a legal RAG system to overcome enterprise LLM limits on private legal knowledge, retrieving evidence to generate grounded responses.
- Built a processing and indexing pipeline for 644 legal documents, creating 161K recursively chunked records with 1024-dimensional BGE-M3 embeddings for scalable semantic retrieval.
- Implemented hybrid retrieval using OpenSearch BM25 and dense vector search with RRF, generating 8–10 candidate contexts before BGE cross-encoder reranking selected the Top-5 contexts for Llama 3.2 3B grounded generation.
- Developed an automated RAGAS evaluation pipeline using GPT-OSS 120B via Groq, achieving 91.4% Faithfulness, 86.7% Answer Relevancy, 82.3% Context Precision, and 79.8% Context Recall

Fraud Claim Detection | *Technologies: Python | Pandas | NumPy | Scikit-learn | Statsmodels | RFECV | GridSearchCV | Matplotlib | Seaborn*

- Built an insurance fraud classifier, prioritizing recall to maximize fraud capture and reduce costly false-negative decisions.
- Collected, analyzed, and engineered insurance claim data for model training through an end-to-end data preprocessing and feature engineering pipeline, addressing missing values and multicollinearity, consolidating low-frequency categorical variables
- Optimized Logistic Regression and Random Forest using RFECV, stratified validation, GridSearchCV, and threshold analysis; selected a 0.40 Logistic Regression cutoff.
- Evaluated with accuracy, precision, recall, F1-score, and confusion matrix; Logistic Regression achieved 77.3% accuracy, 45.9% recall, and 0.500 F1 versus Random Forest’s 25.7% recall and 0.342 F1.

EDUCATION & CERTIFICATIONS

B.Tech in Mechanical Engineering — **Vellore Institute of Technology** | Vellore, Tamil Nadu **Jun 2019 – May 2023**

CERTIFICATIONS

TCS iON Career Edge - Generative AI Essentials certification (2026) | Novi Tech Private Ltd - Artificial Intelligence (2025)